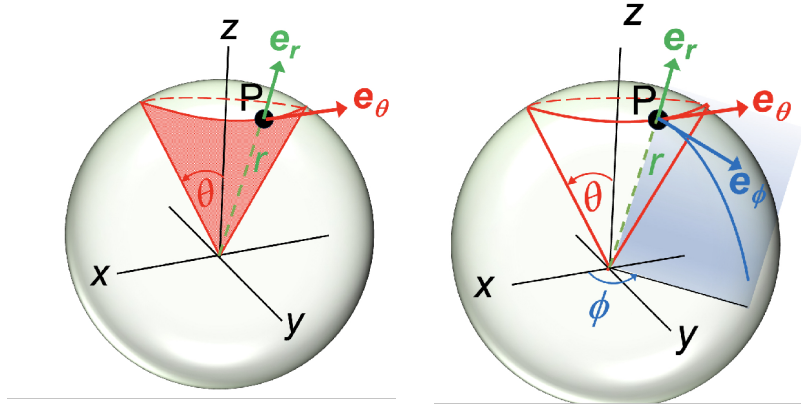


Spherical and cylindrical coordinates in General Relativity (Sections 1.1 and 1.2)

Spherical



The RH figure shows three surfaces that intersect at a point P. The LH figure shows two of those surfaces, a green sphere of radius r about the origin and a red cone around the z -axis at angle θ . The intersection is the circle passing through P. The basis vector $\mathbf{e}_\theta$ at point P is shown tangent to the circle. The basis vector $\mathbf{e}_r$ lies on the line that passes through the origin and P (because the tangent to a line is the line itself). The RH figure adds the half-plane generated by angle ϕ , and the blue arc is where it intersects the sphere. $\mathbf{e}_\phi$ is shown tangent to that arc at P.

The **position vector** is

$$\mathbf{r} = \mathbf{r}(r, \phi, \theta) \stackrel{(1.3)}{=} x \mathbf{i} + y \mathbf{j} + z \mathbf{k} = r \sin \theta \cos \phi \mathbf{i} + r \sin \theta \sin \phi \mathbf{j} + r \cos \theta \mathbf{k}$$

and

$$\begin{aligned} x &= r \sin \theta \cos \phi & y &= r \sin \theta \sin \phi & z &= r \cos \theta \\ r &= \sqrt{x^2 + y^2 + z^2} & \theta &= \arccos \frac{z}{r} & \phi &= \arctan \frac{y}{x} \end{aligned}$$

The **natural basis** is:

$$\begin{aligned} \mathbf{e}_1 = \mathbf{e}_r &= \frac{\partial \mathbf{r}}{\partial r} = \frac{\partial x}{\partial r} \mathbf{i} + \frac{\partial y}{\partial r} \mathbf{j} + \frac{\partial z}{\partial r} \mathbf{k} = \sin \theta \cos \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \theta \mathbf{k} \\ \mathbf{e}_2 = \mathbf{e}_\theta &= \frac{\partial \mathbf{r}}{\partial \theta} = \frac{\partial x}{\partial \theta} \mathbf{i} + \frac{\partial y}{\partial \theta} \mathbf{j} + \frac{\partial z}{\partial \theta} \mathbf{k} = r \cos \theta \cos \phi \mathbf{i} + r \cos \theta \sin \phi \mathbf{j} - r \sin \theta \mathbf{k} \\ \mathbf{e}_3 = \mathbf{e}_\phi &= \frac{\partial \mathbf{r}}{\partial \phi} = \frac{\partial x}{\partial \phi} \mathbf{i} + \frac{\partial y}{\partial \phi} \mathbf{j} + \frac{\partial z}{\partial \phi} \mathbf{k} = -r \sin \theta \sin \phi \mathbf{i} + r \sin \theta \cos \phi \mathbf{j}, \end{aligned}$$

where $r > 0$ and $0 < \theta < \pi$.

The covariant **metric tensor** is

$$G = (g_{ij}) = (\mathbf{e}_i \cdot \mathbf{e}_j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}. \quad \checkmark$$

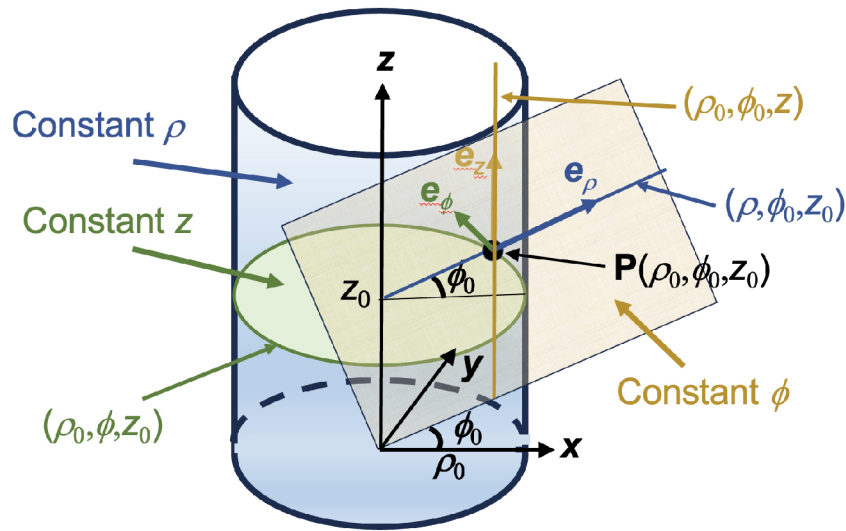
The **dual basis** is:

$$\begin{aligned} \mathbf{e}^1 &= \mathbf{e}^r = \nabla r = \frac{\partial r}{\partial x} \mathbf{i} + \frac{\partial r}{\partial y} \mathbf{j} + \frac{\partial r}{\partial z} \mathbf{k} \\ &= \frac{x}{\sqrt{x^2+y^2+z^2}} \mathbf{i} + \frac{y}{\sqrt{x^2+y^2+z^2}} \mathbf{j} + \frac{z}{\sqrt{x^2+y^2+z^2}} \mathbf{k} \\ &= \sin \theta \cos \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \theta \mathbf{k} \\ \mathbf{e}^2 &= \mathbf{e}^\theta = \nabla \theta = \frac{\partial \theta}{\partial x} \mathbf{i} + \frac{\partial \theta}{\partial y} \mathbf{j} + \frac{\partial \theta}{\partial z} \mathbf{k} \\ &= \frac{xz}{(x^2+y^2+z^2) \sqrt{x^2+y^2}} \mathbf{i} + \frac{yz}{(x^2+y^2+z^2) \sqrt{x^2+y^2}} \mathbf{j} - \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \mathbf{k} \\ &= \frac{\cos \theta \cos \phi}{r} \mathbf{i} + \frac{\cos \theta \sin \phi}{r} \mathbf{j} - \frac{\sin \theta}{r} \mathbf{k} \\ \mathbf{e}^3 &= \mathbf{e}^\phi = \nabla \phi = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} = -\frac{y}{x^2+y^2} \mathbf{i} + \frac{x}{x^2+y^2} \mathbf{j} \\ &= -\frac{\sin \phi}{r \sin \theta} \mathbf{i} + \frac{\cos \phi}{r \sin \theta} \mathbf{j} \end{aligned}$$

The contravariant **metric tensor** is

$$\hat{G} = (g^{ij}) = (\mathbf{e}^i \cdot \mathbf{e}^j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}. \quad \checkmark$$

Cylindrical



The figure shows a cylinder, the surface of constant $\rho = \rho_0$, shaded in blue; the disc of constant $z = z_0$ in green; and the half-plane of constant $\phi = \phi_0$ in tan. The point P is at the intersection of the three surfaces. The boundary of the green disc is a circle having ϕ for its parameter, and, so, the natural basis vector $\mathbf{e}_\phi$ lies along a tangent to the circle at P. The blue line through P is parameterized by ρ , and, so $\mathbf{e}_\rho$ points outward from P as shown. It is on the blue line because the line is the tangent to itself. The vertical tan line through P is parameterized by z , and $\mathbf{e}_z$ points upward from P as shown, tangent to the line. The dual basis vectors (not shown) lie normal to the three surfaces and are parallel to their respective natural basis vectors.

The **position vector** is

$$\mathbf{r} = \mathbf{r}(\rho, \phi, z) \stackrel{(1.3)}{=} x \mathbf{i} + y \mathbf{j} + z \mathbf{k} = \rho \cos \phi \mathbf{i} + \rho \sin \phi \mathbf{j} + z \mathbf{k},$$

where

$$\begin{aligned} x &= \rho \cos \phi & y &= \rho \sin \phi & z &= z \\ \rho &= \sqrt{x^2 + y^2} & \phi &= \arctan\left[\frac{y}{x}\right] & z &= z. \end{aligned}$$

The **natural basis** consists of the tangent vectors of $\mathbf{r}$:

$$\mathbf{e}_\rho = \frac{\partial \mathbf{r}}{\partial \rho} = \frac{\partial x}{\partial \rho} \mathbf{i} + \frac{\partial y}{\partial \rho} \mathbf{j} + \frac{\partial z}{\partial \rho} \mathbf{k} = \cos\phi \mathbf{i} + \sin\phi \mathbf{j}$$

$$\mathbf{e}_\phi = \frac{\partial \mathbf{r}}{\partial \phi} = \frac{\partial x}{\partial \phi} \mathbf{i} + \frac{\partial y}{\partial \phi} \mathbf{j} + \frac{\partial z}{\partial \phi} \mathbf{k} = -\rho \sin\phi \mathbf{i} + \rho \cos\phi \mathbf{j}$$

$$\mathbf{e}_z = \frac{\partial \mathbf{r}}{\partial z} = \frac{\partial x}{\partial z} \mathbf{i} + \frac{\partial y}{\partial z} \mathbf{j} + \frac{\partial z}{\partial z} \mathbf{k} = \mathbf{k},$$

where $\rho > 0$ and $0 \leq \phi < 2\pi$.

The covariant **metric tensor** is $G = (g_{ij}) = (\mathbf{e}_i \cdot \mathbf{e}_j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

To find the **dual basis**, first convert the partial derivatives from x, y, z to ρ, ϕ , and z :

$$\frac{\partial \rho}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \frac{\rho \cos\phi}{\sqrt{\rho^2}} = \cos\phi, \quad \frac{\partial \rho}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \frac{\rho \sin\phi}{\sqrt{\rho^2}} = \sin\phi, \quad \frac{\partial \rho}{\partial z} = 0$$

$$\frac{\partial \phi}{\partial x} = \frac{-y}{x^2+y^2} = \frac{-\rho \sin\phi}{\rho^2} = \frac{-\sin\phi}{\rho}, \quad \frac{\partial \phi}{\partial y} = \frac{x}{x^2+y^2} = \frac{\rho \cos\phi}{\rho^2} = \frac{\cos\phi}{\rho}, \quad \frac{\partial \phi}{\partial z} = 0$$

$$\frac{\partial z}{\partial x} = 0, \quad \frac{\partial z}{\partial y} = 0, \quad \frac{\partial z}{\partial z} = 1$$

The dual basis is composed of the gradient vectors, defined in terms of the partials:

$$\mathbf{e}^\rho = \nabla \rho = \frac{\partial \rho}{\partial x} \mathbf{i} + \frac{\partial \rho}{\partial y} \mathbf{j} + \frac{\partial \rho}{\partial z} \mathbf{k} = \cos\phi \mathbf{i} + \sin\phi \mathbf{j}$$

$$\mathbf{e}^\phi = \nabla \phi = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} = -\frac{\sin\phi}{\rho} \mathbf{i} + \frac{\cos\phi}{\rho} \mathbf{j}$$

$$\mathbf{e}^z = \nabla z = \frac{\partial z}{\partial x} \mathbf{i} + \frac{\partial z}{\partial y} \mathbf{j} + \frac{\partial z}{\partial z} \mathbf{k} = \mathbf{k}$$

Observe further that the corresponding natural and dual basis vectors have the same direction, only differing by up to a scalar factor. Further, observe that the basis vectors are orthogonal. For example, $\mathbf{e}_\rho \cdot \mathbf{e}_\phi = 0$.

The contravariant **metric tensor** is $\hat{G} = (g^{ij}) = (\mathbf{e}^i \cdot \mathbf{e}^j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\rho^2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 1.6.2 Find the line element ds^2 for the following surfaces: (a) Sphere of radius a , (b) Cylinder of radius a , (c) Hyperbolic paraboloid of Example 1.6.1

Solution.

One way to find the line element is to directly use the line element formula for a surface:

$$ds^2 = g_{AB} du^A du^B \text{ for } A, B = 1, 2$$

u^A is known in all three cases:

(a) $u^1 = \theta$ and $u^2 = \phi$

(b) $u^1 = \theta$ and $u^2 = z$

(c) $u^1 = u$ and $u^2 = v$

If we don't already have the metric tensor components, we get it from

$$g_{AB} = \mathbf{e}_A \cdot \mathbf{e}_B$$

by computing

$$\mathbf{e}_A = \frac{\partial \mathbf{r}}{\partial u^A} = \frac{\partial x}{\partial u^A} \mathbf{i} + \frac{\partial y}{\partial u^A} \mathbf{j} + \frac{\partial z}{\partial u^A} \mathbf{k}$$

where

$$\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}.$$

Here we use a direct approach, starting with $\mathbf{r}$, computing $d\mathbf{r}$, and then

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

(a) Sphere of radius a

Fixing $r = a$, then from Example 1.1.4,

$$\mathbf{r} = a \sin \theta \cos \phi \mathbf{i} + a \sin \theta \sin \phi \mathbf{j} + a \cos \theta \mathbf{k}$$

$$d\mathbf{r} = a (\cos \theta \cos \phi d\theta - \sin \theta \sin \phi d\phi)$$

$$+ a (\cos \theta \sin \phi d\theta + \sin \theta \cos \phi d\phi)$$

$$- a \sin \theta d\theta$$

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

$$= a^2 [(\cos^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \sin^2 \theta) d\theta^2$$

$$+ (\sin^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi) d\phi^2]$$

$$= a^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad \checkmark$$

(b) Cylinder of radius a

Fixing $r = a$, then from Exercise 1.1.1,

$$\mathbf{r} = a \cos \phi \mathbf{i} + a \sin \phi \mathbf{j} + z \mathbf{k},$$

$$d\mathbf{r} = -a \sin \phi d\phi \mathbf{i} + a \cos \phi d\phi \mathbf{j} + dz \mathbf{k}$$

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

$$= a^2 (\sin^2 \phi + \cos^2 \phi) d\phi^2 + dz^2 = a^2 d\phi^2 + dz^2 \quad \checkmark$$

(C) Hyperbolic paraboloid

$$\mathbf{r} = (u + v) \mathbf{i} + (u - v) \mathbf{j} + 2uv \mathbf{k}$$

$$d\mathbf{r} = (du + dv) \mathbf{i} + (du - dv) \mathbf{j} + 2(v du + u dv) \mathbf{k}$$

$$ds^2 = d\mathbf{r} \cdot d\mathbf{r}$$

$$= 2(1 + 2v^2) du^2 + 2(1 + 2u^2) dv^2 + 8uv du dv \quad \checkmark$$